# JP MORGAN FINANCIAL RISK ANALYSIS USING PYTHON

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Course: Data Science

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Project Type: End-to-End Financial Risk Analytics Project

Tools Used: Python, Pandas, NumPy, Matplotlib, SciPy

Submission Type: Final Project Summary Report

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# 1. INTRODUCTION

Banks and financial institutions process a very large volume of customer transactions every day. Understanding how customers transact, how money flows across accounts, and where financial risk exists is critical for maintaining portfolio stability, reducing losses, and improving customer engagement.

This project focuses on analyzing customer transaction data using Python to understand transaction behavior, identify high-value and high-risk accounts, detect anomalous and irregular patterns, and support decision-making using data-backed statistical validation.

The analysis follows a structured, task-based approach and converts raw transactional data into actionable business insights that are relevant for real-world banking and financial risk management.

# 2. PROBLEM STATEMENT

The key business questions addressed in this project include:

• Are customers depositing more money or withdrawing more money over time?

• Which accounts are financially strong and which accounts are risky?

• Can early warning signs of financial stress be identified using data?

• Does high transaction activity indicate higher customer value?

• How can data-driven analysis improve monitoring and engagement strategies?

The objective of this project is to support financial risk management and customer profiling using structured data analytics and statistical reasoning.

# 3. DATA UNDERSTANDING

The dataset used in this project contains transaction-level information for customer accounts. It includes details related to customers, accounts, transaction activity, balances, and risk indicators.

Key data elements include:

• Customer and Account identifiers

• Transaction dates and transaction amounts

• Transaction types (Deposit, Withdrawal, Payment, Transfer)

• Account balances and risk-related metrics

## Column Classification

• Identifiers: TransactionID, CustomerID, AccountID

• Categorical: AccountType, TransactionType, Product, Region

• Numerical: TransactionAmount, AccountBalance, RiskScore

• Date: TransactionDate

This dataset supports both behavioral analysis and financial risk assessment, making it suitable for end-to-end analytical evaluation.

## 4. TASK 1 – DATA CLEANING AND FORMATTING

### Objective

The objective of Task 1 was to prepare the raw transactional dataset for analysis by ensuring data accuracy, consistency, and usability. Since all subsequent analysis depends on clean data, this step is critical to the overall success of the project.

### Data Cleaning Steps Performed

• Financial fields were validated and converted into numeric format.

• Mixed date formats were standardized and converted into datetime format.

• Categorical fields were standardized to remove inconsistencies.

### Data Quality Validation

• No missing values were introduced during cleaning.

• All financial fields were successfully converted to numeric types.

• Date fields were converted from text to datetime format.

### Business Impact

Clean and validated data ensures that transaction trends, customer profiles, risk indicators, and statistical tests are reliable. This task establishes a strong foundation for accurate analysis and prevents misleading conclusions in later stages.

## 5. TASK 2 – DESCRIPTIVE TRANSACTIONAL ANALYSIS

### Objective

To analyze money inflow and outflow patterns across time and accounts and identify performance and inactivity risk.

### Assumptions

• Deposit → Credit (Money In)

• Withdrawal, Payment, Transfer → Debit (Money Out)

• Accounts considered dormant if transaction gap ≥ 60 days

### Analysis Performed

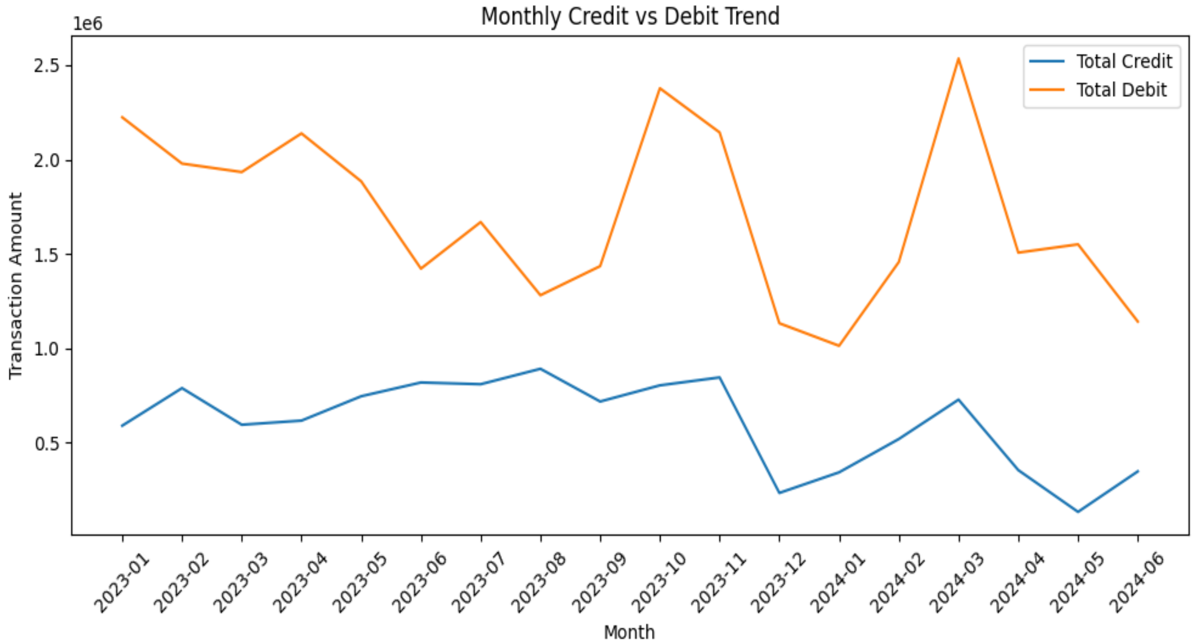
• Calculated monthly and yearly credit, debit, and net transaction volumes

• Visualized credit vs debit trends over time

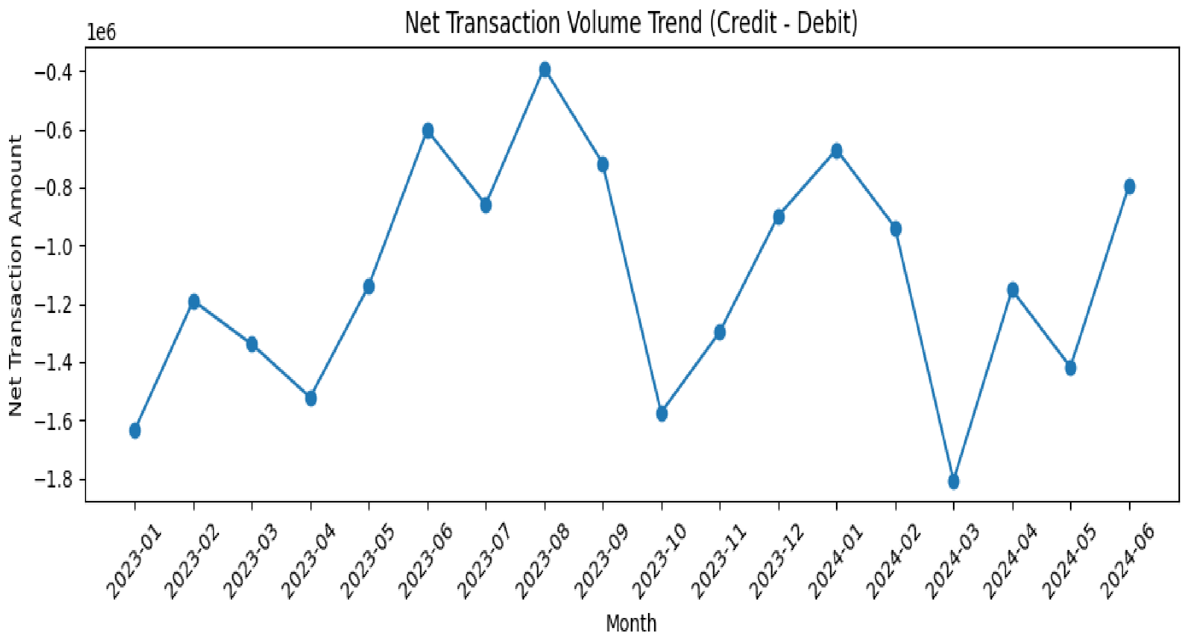
• Identified top and bottom performing accounts

• Flagged dormant accounts

*[Monthly Credit vs Debit Trend Chart]*




*[Net Transaction Trend Over Time Chart]*




## Key Insights

- Debit transactions consistently exceeded credit transactions, leading to net cash outflow
- Certain months showed sharp debit spikes
- A small group of accounts contributed strong positive net inflow
- Several accounts showed prolonged inactivity

# 6. TASK 3 – CUSTOMER PROFILE BUILDING

## Objective

To classify accounts into meaningful segments based on activity, balance behavior, and financial contribution.

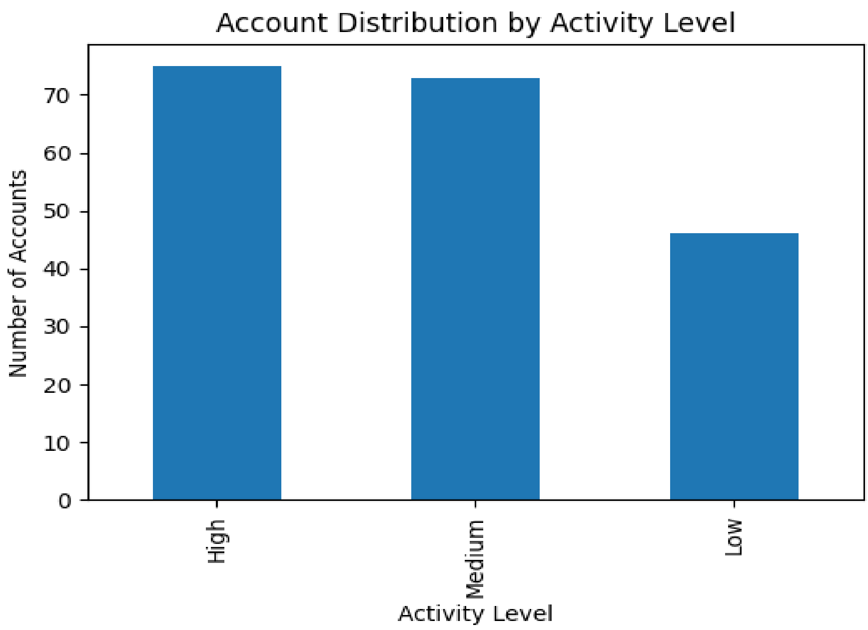
## Activity Level Grouping (Rubric)

• High Activity → Top 33% accounts by transaction frequency

• Medium Activity → Middle 33%

• Low Activity → Bottom 33%

*[ Activity Level Distribution Chart]*




## Special Customer Profiles

• High-Net Inflow Accounts

• High-Frequency Low-Balance Accounts

• Negative or Near-Zero Balance Accounts

## Key Insights

• High activity does not always correspond to high balance

• A small group of accounts contributes disproportionately to net inflow

• High-frequency low-balance accounts indicate potential financial stress

## 7. TASK 4 – FINANCIAL RISK IDENTIFICATION

### Objective

To identify risky and irregular financial behavior using statistical methods.

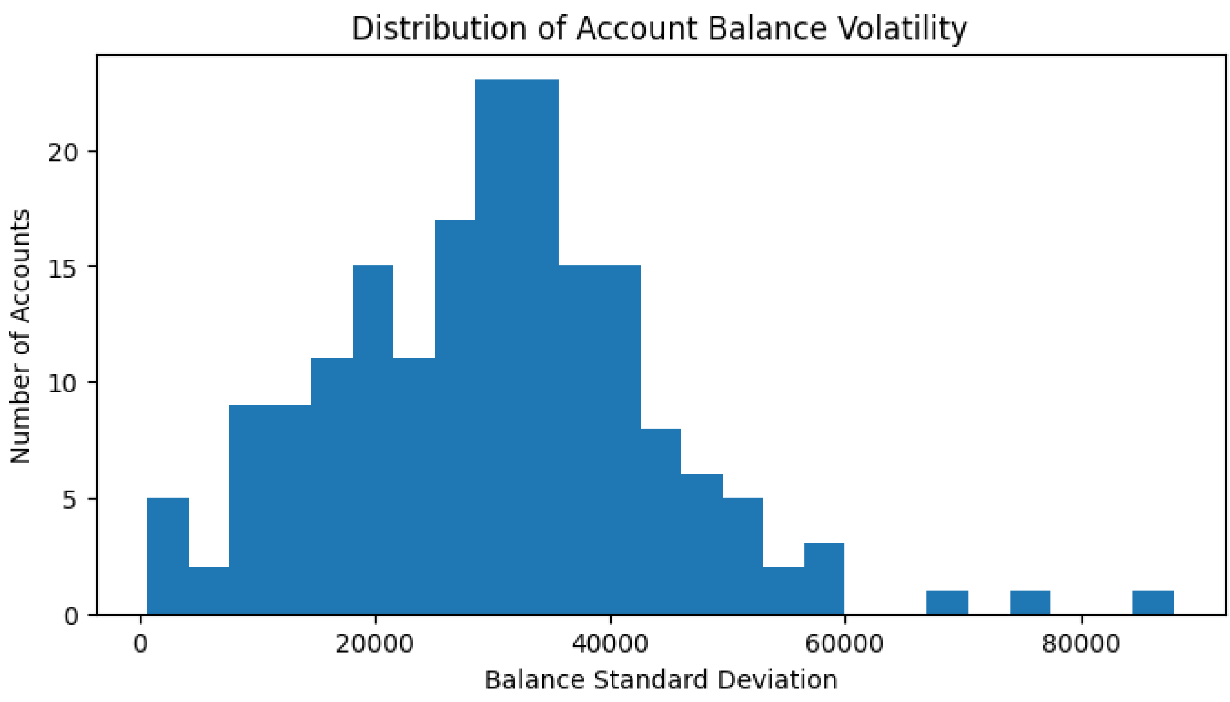
### Risk Detection Approach

• Frequent large withdrawals identified using percentile thresholds

• Balance volatility measured using standard deviation

• Anomalous transactions detected using Z-score

*[ Balance Volatility Distribution Chart]*




### Key Findings

• Some accounts exhibit unusually large withdrawal behavior

• High balance volatility suggests unstable account activity

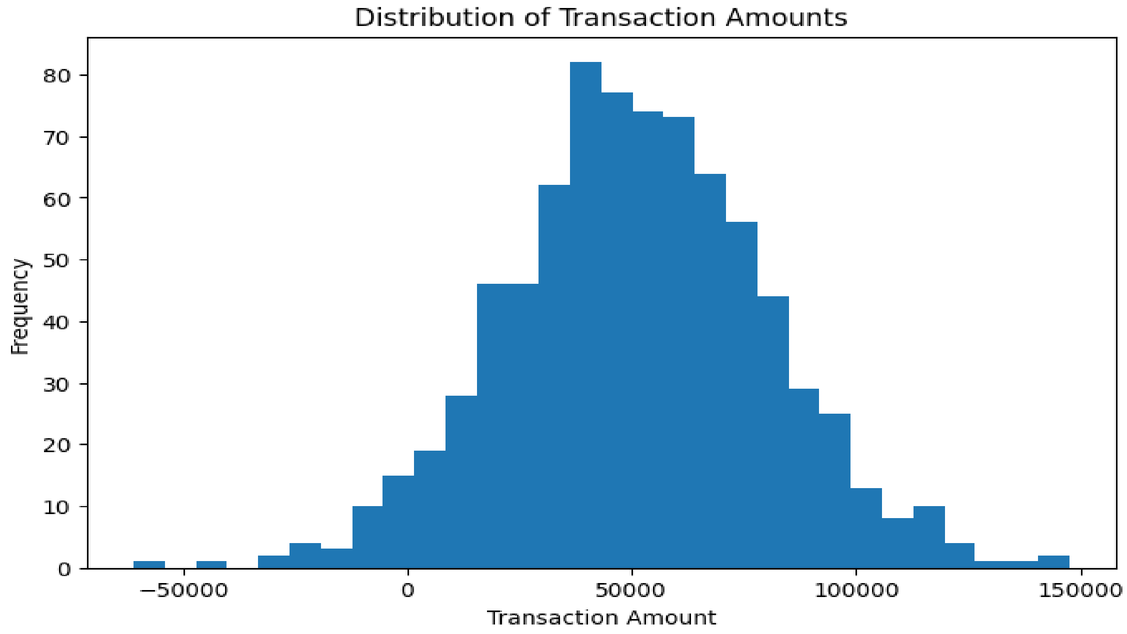
• Extreme transaction amounts identified as outliers

## 8. TASK 5 – EXPLORATORY DATA ANALYSIS & BUSINESS VISUALIZATION

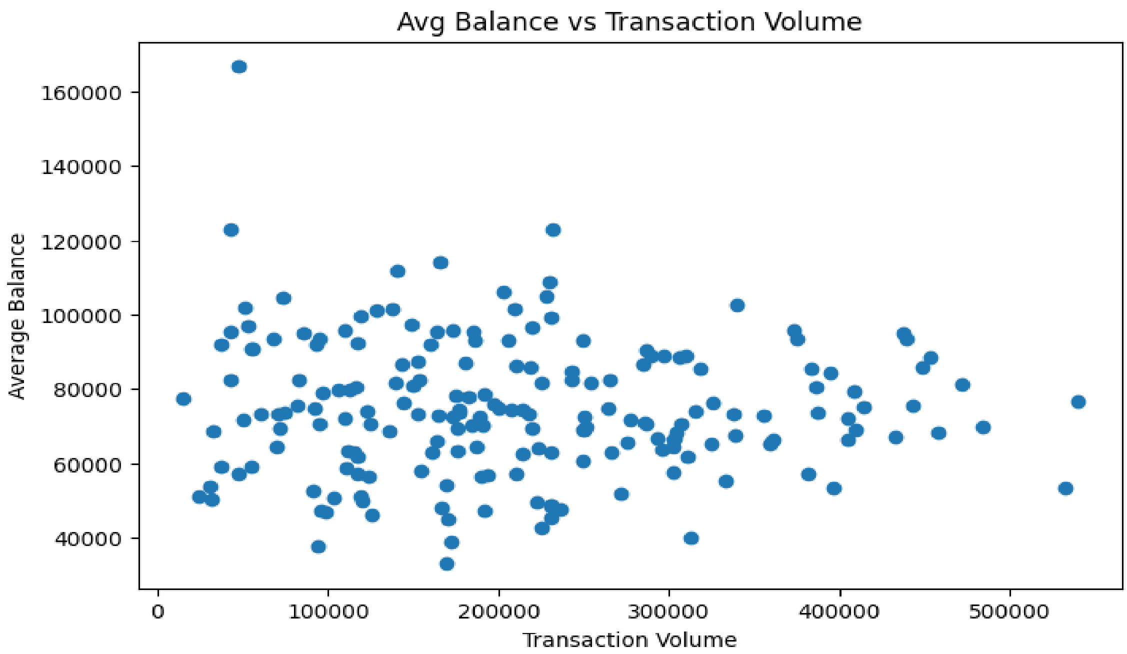
### Objective

To visually analyze transaction behavior, customer engagement, and financial risk patterns.

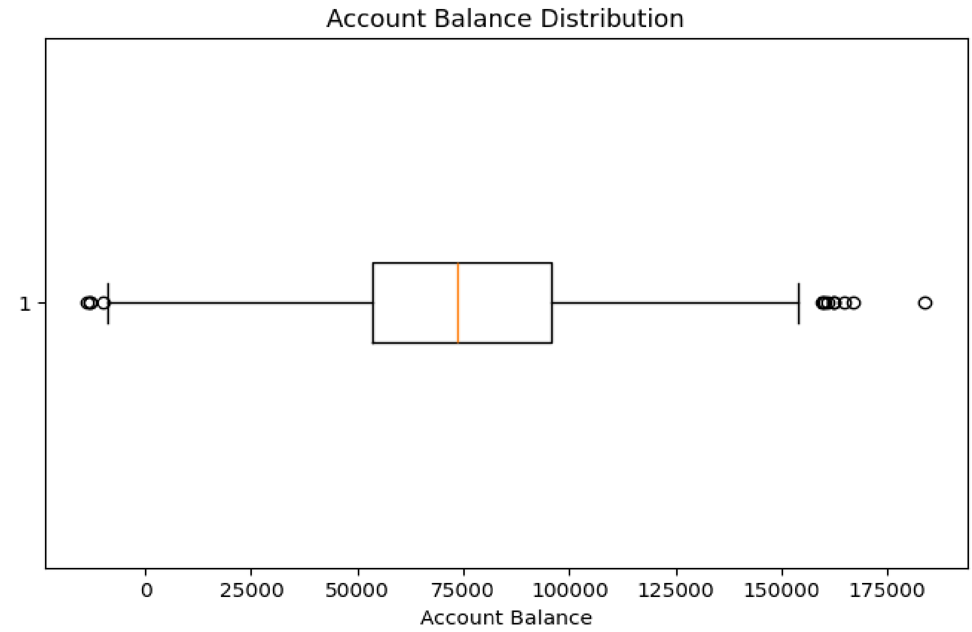
*[Transaction Amount Distribution Chart]*




*[ Balance vs Transaction Volume Chart]*



*[ Account Balance Distribution Chart]*




### Business Insights

• Persistent net cash outflow indicates systemic risk

• Weak relationship between transaction volume and balance

• High engagement alone does not imply financial health

## 9. TASK 6 – HYPOTHESIS TESTING

### Objective

To test whether high-volume transaction accounts maintain higher average balances.

### Methodology

• Accounts segmented using median transaction volume

• Welch's two-sample t-test applied at $\alpha = 0.05$

### Results & Interpretation

• p-value > 0.05

• Null hypothesis not rejected

• No statistically significant difference in average balances

### Business Insight

Transaction activity alone does not guarantee higher financial strength.

## 10. TASK 7 – VIDEO PRESENTATION SUMMARY

A 5-minute video presentation was recorded summarizing project objectives, key findings, and recommendations.

*[LINK - [https://www.loom.com/share/221cbfd770e5409082f6af21ed8d0135](https://www.loom.com/share/221cbfd770e5409082f6af21ed8d0135)]*

## 11. FINAL KEY INSIGHTS

• Debit transactions dominate credit transactions

• High activity does not imply financial stability

• Balance volatility is a strong risk indicator

## 12. BUSINESS RECOMMENDATIONS

• Monitor accounts with high balance volatility and large withdrawals

• Use balance stability and net inflow for customer value assessment

• Apply targeted engagement strategies for high-risk segments

## 13. CONCLUSION

This project demonstrates how Python-based financial analytics can transform raw transaction data into meaningful insights. The structured approach enables effective risk identification, customer profiling, and data-driven decision-making for real-world banking and financial analysis.